

AMENDMENT No. 2 SEPTEMBER 2022
TO
IS 191 : 2007 COPPER — SPECIFICATION
(Fourth Revision)

[Page 4, Table 4 (*see also Amendment No. 1*)] — Substitute the following for the existing:

Amendment No. 2 to IS No. 191 : 2007

Table 4 Chemical Composition
(Unless otherwise indicated all limits are in percentage)
(Clause 7.1)

Designation	Copper + Silver, <i>Min</i>	Silver, <i>Max</i>	Phosphorus	Arsenic, <i>Max</i>	Antimony, <i>Max</i>	Bismuth, <i>Max</i>	Iron <i>Max</i>	Lead, <i>Max</i>	Nickel, <i>Max</i>	Oxygen, <i>Max</i>	Selenium, <i>Max</i>	Sulphur, <i>Max</i>	Tellurium, <i>Max</i>	Tin, <i>Max</i>	Zinc, <i>Max</i>	Selenium + Tellurium, <i>Max</i>	Total Impurities, <i>Max</i>
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Cu-CATH-1	99.99	0.002 5	¹⁾	0.000 5	0.000 4	0.000 1 ³⁾	0.001 0	0.000 5	0.001 0 ⁴⁾	—	0.000 2	0.001 5	0.000 2	^{2) & 5)}	²⁾	0.000 3 (Including bismuth)	0.006 5 (Excluding oxygen)
Cu-CATH-2	99.95	0.007 0	—	0.001 5	0.001 5	0.000 3	0.002 5	0.004 0	0.002 0	—	0.001 0	0.002 5	0.000 5	0.001 0	—	—	0.025 (Excluding oxygen)
Cu-ETP	99.90	—	—	—	—	0.001 0	—	0.005	—	0.045	—	—	—	—	—	—	0.03 (Excluding oxygen and silver)
Cu-FRHC	99.90	—	—	—	—	0.001 0	—	0.005	—	0.045	—	—	—	—	—	—	0.03 (Excluding oxygen and silver)
Cu-FRTP-1	99.85	—	—	0.02	0.005	0.003	0.010	0.005	0.05	0.10	—	—	—	0.01	—	0.025	0.05
Cu- FRTP-2	99.50	—	—	0.10	0.05	0.02	0.03	0.10	0.50	0.10	—	—	—	0.05	—	0.07	—
Cu-DHP	99.80	—	0.015-0.10	0.05	0.005	0.003	0.03	0.01	0.10	—	—	—	0.01	0.01	—	0.02	0.06 (Excluding silver, nickel, arsenic and phosphorus)
Cu-ATP	99.20	—	—	0.20-0.50	0.01	0.005	0.02	0.02	0.15	0.10	—	—	—	0.03	—	0.03	—
Cu-DPA	99.20	—	0.015-0.10	0.20-0.50	0.01	0.003	0.03	0.010	0.15	—	—	—	0.01	0.01	—	0.02	0.06 (Excluding silver, nickel, arsenic and phosphorus)

NOTES

1 (As + Cd + Cr + Mn + P + Sb) maximum 0.001 5.

2 (Co + Fe + Ni + Si + Sn + Zn) maximum 0.002 0.

3 Can be supplied with any other limits of bismuth (with maximum of 0.000 2) as per the agreement between purchaser and supplier.

4 Can be supplied without specifying the limits of nickel as per the agreement between purchaser and supplier.

5 If required by the purchaser only and in such case, the limits shall be as per the agreement between purchaser and the supplier subject to the maximum limit of 0.000 5.

(MTD 08)

Publication Unit, BIS, New Delhi, India